

Employment Outcome Intelligence Platform

A Trusted, Automated, Longitudinal Outcome Verification & Skill Intelligence Layer Connecting Skilling Records to Independently Verified Employer Trajectories.

TRUST

P1: INDEPENDENT VERIFICATION

TRAJECTORY

LONGITUDINAL OUTCOME LAYER

INTELLIGENCE

STATISTICAL LEAKAGE & DEFICITS

LIVE PRODUCTION DEPLOYMENT & ARTIFACTS

Live Web App: <https://eoi-platform.vercel.app>

GitHub Codebase: <https://github.com/Pratik-techh/eoi-platform>

Interactive Demo: <https://eoi-platform.vercel.app/demo>

Audit Ledger: 502 Chained SHA-256 Blocks

Document Version: 2.0 (Complete System Manual)

Engineering Standards: Zero Single-Stakeholder Truth · DPDP Act 2023 Compliant

Status: Evaluator Production Ready

The Problem: The "Trained vs Employed" Discrepancy

Historically, government skilling programs in India (such as PMKVY, DDU-GKY, and NAPS) measured success by asking: *"How many people did we train?"*. Audits by the Comptroller and Auditor General (CAG) repeatedly revealed severe placement discrepancies: training agencies frequently self-reported inflated placement figures, many of which represented short-term, ghost, or fictitious appointments.

The **Employment Outcome Intelligence (EOI) Platform** fundamentally transforms this model. The system shifts policy from simple enrollment counts to answering: *"What happened to them? Who independently verified it? How long did they retain their jobs? And what skills does industry demand that training failed to provide?"*

Core Platform Pitch: "We don't just track who was trained. We track what happened next, verify it independently, preserve it cryptographically, and learn from it."

Non-Negotiable Database Invariants (P1 – P5)

These core principles are enforced directly at the database engine and server proxy level—never purely through UI assumptions:

P1 Zero Single-Stakeholder Truth

No single entity controls truth. A training agency can report a placement claim, but is strictly prohibited from verifying its own claim. Verification requires authenticated employer confirmation.

P2 No Master Administrator

No superuser can unilaterally create, alter, or override critical outcome records. Sensitive actions (e.g. activating scoring versions, closing high-severity investigations) require dual-key multi-party approvals.

P3 Automated Routine Verification

Statistical leakage detection, entity resolution, job readiness band calculation, and audit chaining are 100% automated by database triggers and math routines, freeing humans for governance.

P4 & P5 Cryptographic Ledger & AI Guardrails

Outcome events are append-only and SHA-256 hash-chained. Destructive updates are forbidden. AI acts strictly as an analytical interpreter over aggregated views and cannot verify or fabricate records.

Longitudinal Outcome Life Cycle



02. Roles & Access Control Matrix

Granular Role Segregation

To prevent collusion, bribery, or administrative falsification, the platform enforces 8 fine-grained persona roles grouped across 4 primary stakeholder families. Every API endpoint and server action validates identity, role, ownership, and current entity state.

Role Identifier	Stakeholder Family	Authorized Capabilities	Strictly Prohibited Actions
gov_analyst	Government	Read aggregated national funnels, regional leakage, skill gaps, run policy simulations, query AI analyst.	Cannot view raw unmasked PII; cannot edit or create outcome records.
gov_program_admin	Government	Configure national programs (PMKVY, NAPS), propose metric versions, view agency rankings.	Cannot activate scoring versions alone (requires dual-key authorization).
gov_auditor	Government	Inspect complete cryptographic audit ledger, run live SHA-256 chain verification, triage leakage.	Read-only auditor access; cannot modify ledger records or overwrite history.
agency_officer	Training Agency	Onboard student cohorts, record assessment scores, compute readiness bands, submit placement reports.	Strictly cannot verify own placement reports; cannot view other agencies.
agency_admin	Training Agency	Manage agency training centers, courses, instructor assignments, batch CSV uploads.	Cannot verify employment; cannot alter historical assessment logs.
employer_verifier	Employer	Confirm, Reject, or Request Correction on placement claims for own canonical organization.	Cannot verify other employers' candidates; cannot modify student training history.
employer_admin	Employer	Manage authorized verifiers, register corporate GSTIN/CIN, submit skill deficit feedback.	Cannot verify without verifier delegation; cannot rewrite student profiles.
student	Student / Trainee	View Tamper-Evident Employability Passport, report unemployment/departures, raise disputes.	Cannot verify own employment; cannot edit agency assessment scores.

Dual-Key Multi-Party Authorization Engine

In compliance with Invariant P2 (No Master Administrator), sensitive operations cannot be executed by any single user. They are staged in the `authorization_requests` queue and require independent approval:

Operations Mandating Multi-Party Dual-Key Approval:

- **Activating a New Metric/Scoring Version:** A Program Admin can propose a scoring algorithm revision, but a second Government Officer must review mathematical implications and approve before activation.
- **Overriding Organization Claim Discrepancies:** Resolving conflicting corporate CIN/GSTIN registrations requires dual authorization.
- **Granting Privileged Platform Roles:** Elevating user privileges requires two distinct system administrators.
- **Closing High-Severity Anomaly Investigations:** Fraud flags with $|z| > 2.5$ require multi-stakeholder closure with documented evidence.

Comprehensive National Intelligence Architecture

The Government Command Center provides macro-level outcome intelligence, automated leakage detection, and scenario simulation without exposing vulnerable student PII.

 National KPI Dashboard (/gov/dashboard)

Displays real-time aggregated metrics across all active skilling schemes: 500 Enrolled Trainees, 82.4% Job Readiness, 310 Independently Verified Employed (62.0% true conversion rate), 3-Month Retention Rate (74.2%), and active leakage alerts.

 Outcome Conversion Funnel (/gov/funnel)

Visualizes pipeline drop-offs at every stage: Enrolled (500) → Assessed (460) → Job Ready (412) → Placement Reported (350) → **Verified Employed (310)** → 3-Month Retention (230). Eliminates phantom placement claims.

 Statistical Leakage Engine (/gov/leakage)

Computes statistical Z-scores across all regions. An absolute z-score `|z| ≥ 2.0` flags severe statistical anomaly. **Planted Benchmark:** Detects Rajasthan IT-BPO placement drop-off at `z = 2.94` (high leakage probability).

 Regional & State Heatmaps (/gov/regions)

Compares verification rates and retention longevity across Indian states and districts, highlighting geographical disparities between industrial corridors and remote regions.

 Skill Demand & Deficit (/gov/skills)

Cross-examines skilling course supply against live employer vacancy requirements. Directly identifies critical deficit skills (e.g. Cloud Security, EV Powertrain Diagnostics, Industrial IoT) and flags obsolete curriculums.

 Policy Scenario Simulator (/gov/simulator)

Interactive policy modeling tool. Adjust training capacity sliders, student stipend allocations, and employer partnership subsidies to predict 12-month verified placement yields and government ROI.

 SHA-256 Audit Ledger (/gov/audit)

Inspects the 502 append-only SHA-256 chained event ledger. Features a **Live Chain Integrity Verifier** button that recalculates block hashes sequentially to prove zero data tampering.

 AI Policy Analyst (/gov/ai)

Natural language policy Q&A assistant. Every response cites specific database view rows (e.g. `v_national_funnel`, `v_regional_leakage`). Features automatic deterministic fallback if LLM keys are absent.

CAG Audit Discrepancy Reconciliation: Traditional dashboards report 350 placed students (70%). The EOI Platform proves that only 310 are verified (62%), exposing 40 unverified or fictitious claims instantly.

04. Operational Portals: Agency & Employer

INTAKE & VERIFICATION

Training Agency Operations (`/agency/...`)

The Training Agency portal serves as the operational onboarding, assessment, and reporting layer. It provides agencies with tools to track students while strictly preventing fraudulent self-verification.

Student Intake & CSV

Onboard trainees individually or via bulk CSV import. Validates Aadhaar hash, contact details, and socioeconomic classification.

Assessment Scoring

Log continuous assessments: technical exam marks, practical lab evaluations, soft skills ratings, and mandatory attendance metrics.

Readiness Bands

Engine automatically assigns **High** ($\geq 75\%$), **Medium** (50-74%), or **Low** ($< 50\%$) job readiness bands based on weighted performance.

Employment Placement Reporting Workflow (`/agency/employment/report`)

When a trainee secures employment, the agency submits a placement report specifying: Candidate Name, Canonical Employer (from registry), Job Designation, Monthly Salary, and mandatory proof documents (Offer Letter / Salary Slip PDF).

 **P1 Invariant Guard:** Upon submission, the record status moves to `reported_unverified`. The agency's interface displays "Awaiting Independent Employer Confirmation". The agency cannot approve this claim.

Employer Verification Queue (`/employer/...`)

The Employer Portal acts as the independent ground truth anchor. Only verified corporate officers with authenticated domain accounts can act on pending claims.

1. Confirm Employment

Employer validates candidate identity, start date, and salary bracket. Transitions record to `verified_employed`, generates a cryptographic event block, and activates the 3-month retention tracking clock.

2. Request Correction

If designation or joining date has a minor typo, employer requests correction. Sends the record back to the agency's correction inbox with specific feedback notes.

3. Reject Placement Claim

If candidate was never hired or ghosted, employer clicks Reject with a mandatory reason code. Creates an anomaly event in the auditor ledger and flags agency.

Departure & Unemployment Reconciliation

Employers can formally confirm when an employee resigns or departs. Records official separation reason and closes the active retention period.

Corporate Skill Deficit Feedback (`/employer/feedback`)

Direct structured channel for employers to report curriculum deficiencies directly to MSDE (e.g. "Candidates excel in Python syntax but lack Git collaboration skills").

Student Trainee Portal (/student/...)

Trainees own their career trajectory and verifiable credentials. The portal empowers students to demonstrate verified employability without relying on paper certificates.

 **Verifiable Employability Passport (** `/student/passport` **)**

Tamper-evident digital passport containing verified NSQF course credentials, assessed technical skills, soft skill evaluations, and cryptographic employer verification stamps. Includes an instant printable PDF export.

 **Longitudinal Career Trajectory (** `/student/history` **)**

Visual timeline mapping the learner's journey across training, assessment, placement, verification, 3-month retention, departures, and subsequent re-employment.

 **Unemployment & Departure Self-Reporting**

If a student leaves a job or faces contract termination, they can self-report departure directly. Triggers automatic re-skilling recommendations and job portal matching.

 **Dispute Resolution Center (** `/student/disputes` **)**

Allows trainees to contest inaccurate agency assessment marks or fraudulent employer placement reports with document uploads and grievance tracking.

Technical & Cryptographic Architecture

Frontend Framework	Next.js 16.3.6 App Router (54 routes), TypeScript strict, Tailwind CSS 4, custom design tokens.
Database Architecture	Dual-mode engine: Full Supabase PostgreSQL migrations (001–011) + standalone deterministic fallback engine (<code>lib/db/store.ts</code>) for zero-setup offline demos.
Cryptographic Ledger	Append-only SHA-256 hash chaining: <code>hash = SHA256(prev_hash + event_type + payload + timestamp + actor_id)</code> . Immutable block storage.
Zero-Trust Middleware	Server-side authorization check (<code>lib/authorize.ts</code>) mirrored with DB Row-Level Security (RLS) policies. Role escalation automatically blocked.
DPDP Act 2023 Compliance	Aadhaar numbers hashed irreversibly; raw PII masked in all analyst/auditor views; consent-driven visibility controls for student profiles.

Evaluator Quick Start & Demo Credentials

Use the 1-click persona switchers on the live site at <https://eoi-platform.vercel.app/demo> or enter credentials:

Persona Role	Demo Login Email	Universal Password	Landing Screen
Government Analyst	<code>gov.analyst@eoi.demo</code>	<code>Demo@EOI2026</code>	<code>/gov/dashboard</code>
Training Agency Officer	<code>agency.officer@eoi.demo</code>	<code>Demo@EOI2026</code>	<code>/agency/dashboard</code>
Employer Verifier	<code>employer.verifier@eoi.demo</code>	<code>Demo@EOI2026</code>	<code>/employer/verification</code>
Student Trainee	<code>student.x@eoi.demo</code>	<code>Demo@EOI2026</code>	<code>/student/dashboard</code>
Government Auditor	<code>gov.auditor@eoi.demo</code>	<code>Demo@EOI2026</code>	<code>/gov/audit</code>